

Energy supplied = $(1.0086 + 1.0097 - 2.0150) \text{ u c}^{-2} / (1.6 \times 10^{-19}) = 3 \text{ MeV}$

30 Ans: C

When the source is first used:

Total number of disintegrations = $A_0 \times 10 \text{ mins}$ --- (1)

2 years later:

$$A_2 = \left(\frac{1}{2}\right)^{2/4} A_0 = \left(\frac{1}{2}\right)^{0.5} A_0 = 0.7071 A_0$$

Total number of disintegrations = $A_2 \times t$

Total number of disintegrations = $0.7071 A_0 \times t$ --- (2)

(2) = (1):

$$0.7071 A_0 \times t = A_0 \times 10 \text{ mins}$$

$$t = 14.1 \text{ mins}$$